Compute Community Presence-Absence Matrix

Description

Computes a binary presence-absence matrix from numeric community proportions or counts. Positive values become '1L'; zeros and missing values become '0L'.

Usage

```
compute_community_presence_absence(mat_community = NULL)
```

Arguments

`mat_community`	A numeric matrix with samples as rows and taxa as columns, as returned by 'prepare_community_for_fit()'. Values must be non-negative.

Details

Presence-absence conversion before 'filter_constant_taxa()' is essential when using a binomial error family: a taxon recorded at non-zero varying proportions in every sample has positive SD on the proportion scale but becomes a constant-1 column after conversion inside the model, causing implicit intercept saturation. Applying this function first ensures that 'filter_constant_taxa()' removes such taxa before they reach the model.

For other error families (e.g., Gaussian, future hurdle models) the raw proportional matrix should be passed to 'filter_constant_taxa()' directly; use the 'error_family' configuration key to control this choice in the pipeline.

Value

An integer matrix with the same dimensions and dimnames as 'mat_community', containing only '0L' and '1L'.

See Also

[prepare_community_for_fit()], [filter_constant_taxa()], [build_jsdm_fit_input()]

Examples

```
mat_community <-

  base::matrix(

    base::c(0, 0.25, NA_real_, 1),

    nrow = 2
```

```
  )


compute_community_presence_absence(
  mat_community = mat_community
)
```